

Literature Benchmark Brief: Pierre_2019_TPWRS

Title: Design of the Pacific Northwest National Laboratory wide-area damping controller

Authors: B. J. Pierre, D. Trudnowski, M. K. Donnelly, V. C. Nguyen, J. F. Gronquist (2019)

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Cluster: Cluster B: SCADA Degradation & Reliability

Core Theoretical Mechanism & Model Formulation

Wide-area PMU latency monitoring, communication watchdog state-machine, bumpless fallback

Empirical Scale & Experimental Dataset

Western Interconnection closed-loop grid testing

Relevance to IEEE TSTE & Sustainable Energy Systems

Defines utility best practice: monitor latency and force graceful fallback to safe baselines

Identified Gaps & Operational Vulnerabilities

Focuses on PMU millisecond scale; does not model wind farm SCADA minute-scale queue delays

Role in Manuscript Defense & Reviewer #2 Proof

Direct justification for watchdog latency monitoring and selective abstention envelope